

Week 2 — In-Class Exercise

Triage the dataset • project a panel by hand • build a cohort by hand

Applied Data Science and AI in Pharma Commercial • Columbia University • Fall 2026

Groups of 3–4 • 25 minutes • one group presents each part in the debrief

Monday, September 21, 2026

Everything in this handout is fictional and synthetic: GLUCOREN is an invented brand, the pharmacies and patients are invented, and the numbers are chosen to be easy to compute by hand. The methods are exactly the ones used on real data.

Part A • Dataset triage (8 min)

Six questions a GLUCOREN brand team asked this year. For each, write: (1) the dataset category or categories you would use (from the nine-category taxonomy), (2) the grain, (3) open or closed, cross-sectional or longitudinal, and (4) the blind spot you would flag in the first sentence of your answer.

#	Question from the brand team	Dataset(s) • grain • open/closed • long./cross-sect. • blind spot
1	"How many type 2 diabetes patients in the Northeast started a branded oral agent last quarter?"	
2	"Which 200 endocrinologists should the new sales team call first?"	
3	"Are patients on GLUCOREN more persistent at 12 months than patients on CARDIVEX?"	
4	"Did the March formulary win with a large PBM increase our share?"	
5	"What share of newly diagnosed DLBCL patients receive CAR-T in second line, and where?"	
6	"How much GLUCOREN did we sell last week, by territory?"	

Part B • Projection by hand (8 min)

A data vendor's panel covers part of GLUCOREN's market. The vendor knows the universe volume of every stratum from wholesaler data. Fill in the coverage, the projection factor and the projected GLUCOREN TRx for each stratum, then the total.

Stratum	Panel pharmacies	Universe pharmacies	Panel weekly volume (all TRx)	Universe weekly volume	Coverage	Factor	Panel GLUCOREN TRx	Projected GLUCOREN TRx
Retail • ZIP3 100	45	70	21,600	31,700			690	
Retail • ZIP3 070	29	51	16,300	26,200			410	
Specialty • ZIP3 070	1	4	330	1,200			5	
Total							1105	

B1. Which stratum's projected number would you trust least, and why?

B2. The twist. Next week the largest chain in Retail · ZIP3 100 — 12 stores, 7,200 weekly TRx, 240 GLUCOREN TRx — leaves the panel. The real market is unchanged. Recompute the stratum's coverage, factor and projection, and the total. (a) What does the raw panel count say happened to GLUCOREN? (b) What does the projected number say? (c) Under what condition is the projected number still biased even after re-weighting?

Space for working:

Part C · Cohort by hand (9 min)

Ten patients from a de-identified extract. Apply tonight's cohort template: ICD-10 C83.30; ≥ 2 claims on different days with the first two ≤ 365 days apart; index = first qualifying day; baseline = continuously enrolled for ≥ 180 days before index; follow-up = censor at coverage end (extract ends 2026-06-30). For each patient write: in / out, the reason, the index date, and whether they are censored before the end of the extract.

Pt	Diagnosis claims (date · ICD-10)	Coverage spells	In / out · reason · index · censored?
1	2025-02-03 · C83.30 2025-02-17 · C83.30 2025-03-10 · C83.30	2023-01-01 → 2026-06-30	
2	2025-04-11 · C83.30	2022-06-01 → 2026-06-30	
3	2025-01-20 · C83.30 2025-02-05 · C83.30	2024-12-01 → 2026-06-30	
4	2023-03-02 · C83.30 2025-05-15 · C83.30 2025-06-01 · C83.30	2022-01-01 → 2026-06-30	
5	2025-06-09 · C82.10 2025-06-30 · C82.10 2025-07-21 · C83.30	2021-01-01 → 2026-06-30	
6	2025-03-03 · C83.30 2025-03-03 · C83.30 2025-03-03 · C83.30	2020-01-01 → 2026-06-30	
7	2025-08-12 · C83.30 2025-08-26 · C83.30	2024-01-01 → 2025-11-30	
8	2025-05-05 · C83.30 2025-05-19 · C83.30	2024-09-01 → 2025-01-31 2025-04-15 → 2026-06-30	
9	2025-02-14 · R59.0 2025-03-01 · C83.30 2025-03-15 · C83.30	2023-05-01 → 2026-06-30	
10	2024-11-04 · C83.30 2025-11-10 · C83.30	2023-01-01 → 2026-06-30	

- C1.** How many patients are in the final cohort? Which single patient is the most debatable call, and what extra data would settle it?
- C2. Re-identification.** Patient 7's row in the de-identified extract contains the fields below. Which combination of fields would let a motivated person (say, a reporter with a local newspaper archive) identify her? Which fields would you suppress or generalize before sharing this table outside the analytics team, and what would that cost the analysis?

Field	Value
patient token	7f3a...9c
birth year	1937
sex	F
ZIP3	121
state	NY
index date	2025-08-12
CAR-T infusion date	2025-12-02
treating NPI	1234567893 (one of two CAR-T centers in the ZIP3)
payer	Medicare